

Decidability In Class Examples and Extra Problems

1. Consider the following language:

$$A_{DFA} = \{\langle B, s \rangle \mid B \text{ is a DFA that accepts input string } s\}$$

Show that A_{DFA} is a decidable language.

2. Consider the following language:

$$A_{NFA} = \{\langle B, s \rangle \mid B \text{ is a NFA that accepts input string } s\}$$

Show that A_{NFA} is a decidable language.

3. Consider the following language:

$$A_{REG} = \{\langle R, s \rangle \mid R \text{ is a regular expression that generates string } s\}$$

Show that A_{REG} is a decidable language.

4. Consider the following language that determines emptiness:

$$E_{DFA} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) = \emptyset\}$$

Show that E_{DFA} is a decidable language.

5. Consider the following language that determines if two languages are equivalent (equal):

$$EQ_{DFA} = \{\langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B)\}$$

Show that EQ_{DFA} is a decidable language.

6. Consider the following language that determines if a context-free grammar generates a particular string:

$$A_{CFG} = \{\langle G, s \rangle \mid G \text{ is a CFG that generates the string } s\}$$

Show that A_{CFG} is a decidable language.

7. Consider the following language that determines if a context-free grammar generates an empty language:

$$E_{CFG} = \{\langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset\}$$

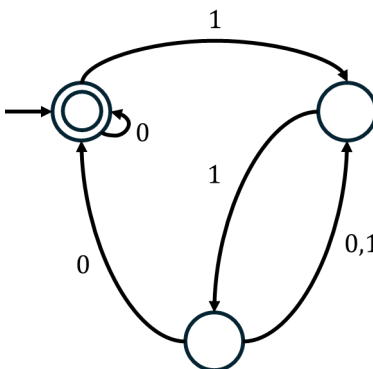
Show that E_{CFG} is a decidable language.

8. Let $ALL_{DFA} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) = \Sigma^*\}$. Show that ALL_{DFA} is a decidable.
9. Let $A\lambda_{CFG} = \{\langle G \rangle \mid G \text{ is a CFG that generates } \lambda\}$. Show that $A\lambda_{CFG}$ is a decidable.
10. Let $E_{TM} = \{\langle M \rangle \mid M \text{ is a Turing Machine and } L(M) = \emptyset\}$. Show that $\overline{E_{TM}}$ is Turing-recognizable.
11. Let $INFINITE_{DFA} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) \text{ is an infinite language}\}$. Show that $INFINITE_{DFA}$ is decidable.
12. Let $A = \{\langle M \rangle \mid M \text{ is a DFA that doesn't accept any string containing an odd number of 1s}\}$. Show that A is decidable.
13. Let $\Sigma = \{0, 1\}$. Show that the problem of determining whether a CFG generates some string in 1^* is decidable. In other words, show that

$$\{\langle G \rangle \mid G \text{ is a CFG over } \{0, 1\} \text{ and } 1^* \cap L(G) \neq \emptyset\}$$

is a decidable language.

14. Answer all parts for the following DFA M and give reasons for your answers.



- (a) Is $\langle M, 0100 \rangle \in A_{DFA}$?
- (b) Is $\langle M, 011 \rangle \in A_{DFA}$?
- (c) Is $\langle M \rangle \in A_{DFA}$?
- (d) Is $\langle M, 0100 \rangle \in A_{REX}$?
- (e) Is $\langle M \rangle \in E_{DFA}$?
- (f) Is $\langle M, M \rangle \in EQ_{DFA}$?